

Suggested FYP topics for 2026-27

by

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Project code:	WJY1
Project title:	Multi-Modal Trajectory Generation and Masked Action Planning in World Models for Autonomous Driving
No. of students:	1~2
Description of project:	Autonomous driving systems must safely navigate complex traffic by predicting multiple plausible future trajectories. Traditional end-to-end planners often rely on handcrafted trajectory anchors, which can discretize the solution space and struggle to capture the full diversity of natural driving behaviors. This project explores an innovative autonomous driving framework that formulates path planning as a masked sequence completion task using a world model. Students will investigate how to generate diverse, history-consistent trajectory modes from latent states without relying on predefined anchor sets. Furthermore, the project involves integrating a lightweight world model to predict future Bird's-Eye-View (BEV) semantics conditioned on these generated paths. By doing so, the model learns from a full distribution of plausible futures, aiming to achieve state-of-the-art safety and trajectory accuracy while maintaining the real-time inference latency required for physical deployment.
Industrial collaboration :	No.

Project code:	WJY 2
Project title:	Accelerating Vision-Based UAV Path Planning using State-Space World Models
No. of students:	1~2
Description of project:	Vision-based path planning for Unmanned Aerial Vehicles (UAVs) involves highly non-linear dynamics and noisy sensor signals, making standard reinforcement learning highly data-inefficient. Model-based reinforcement learning (MBRL) solves this by learning a "world model" of the environment's dynamics, though traditional recurrent world models suffer from slow, computationally expensive sequential training. This project will investigate the application of modern, parallelizable State-Space Models (SSMs) to accelerate the training of world models for agile quadrotor flight. Students will develop a framework that handles long temporal sequences efficiently while optionally utilizing privileged state information in simulation to overcome the challenges of partial observability from raw visual inputs. The ultimate goal is to train a robust, high-speed UAV control policy capable of navigating complex tracks with significantly reduced training times and a minimized sim-to-real gap.

Industrial collaboration:	No.
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